Berka Database

Part 1: Database Exploring

1. Summary

This project explores the Berka database, a real-world relational banking dataset from a Czech bank. It covers the full operational structure of the bank from 1993 to 1998 with 4,500 accounts and 1.06 million transactions across eight tables.

This project delivers two things:

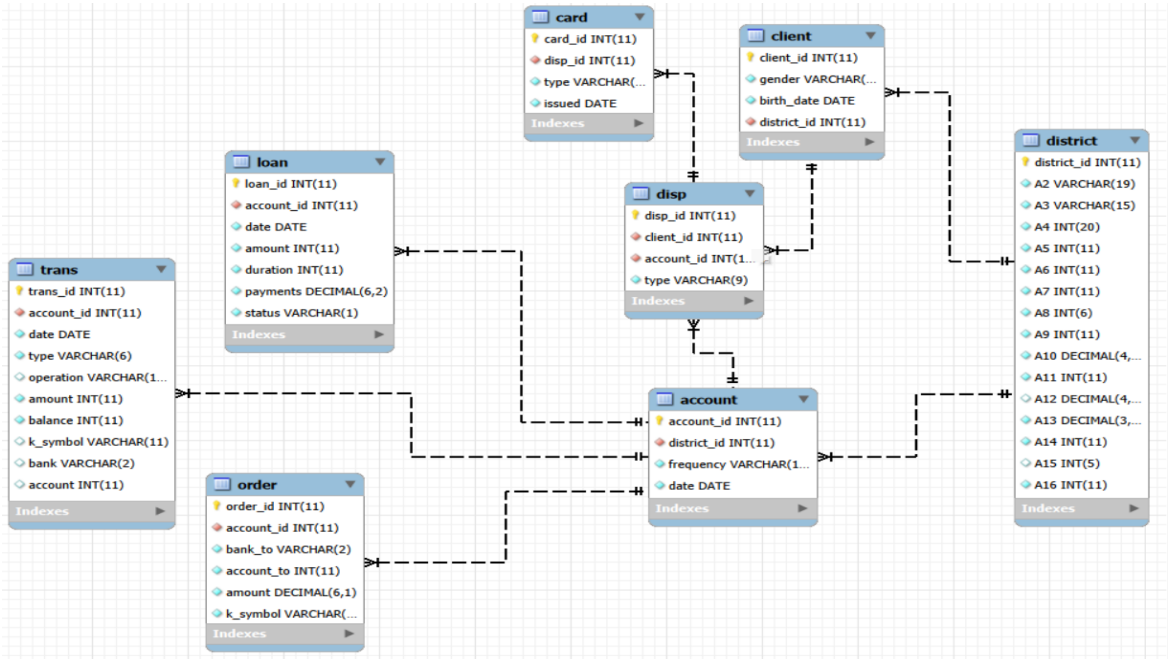
- **An interactive visualization** to explore the bank by region, age, year, fees, orders, loans and income.
- **A bank-level money-flow visualization** showing the monthly average flow of money.

2. Data Overview and Methodology

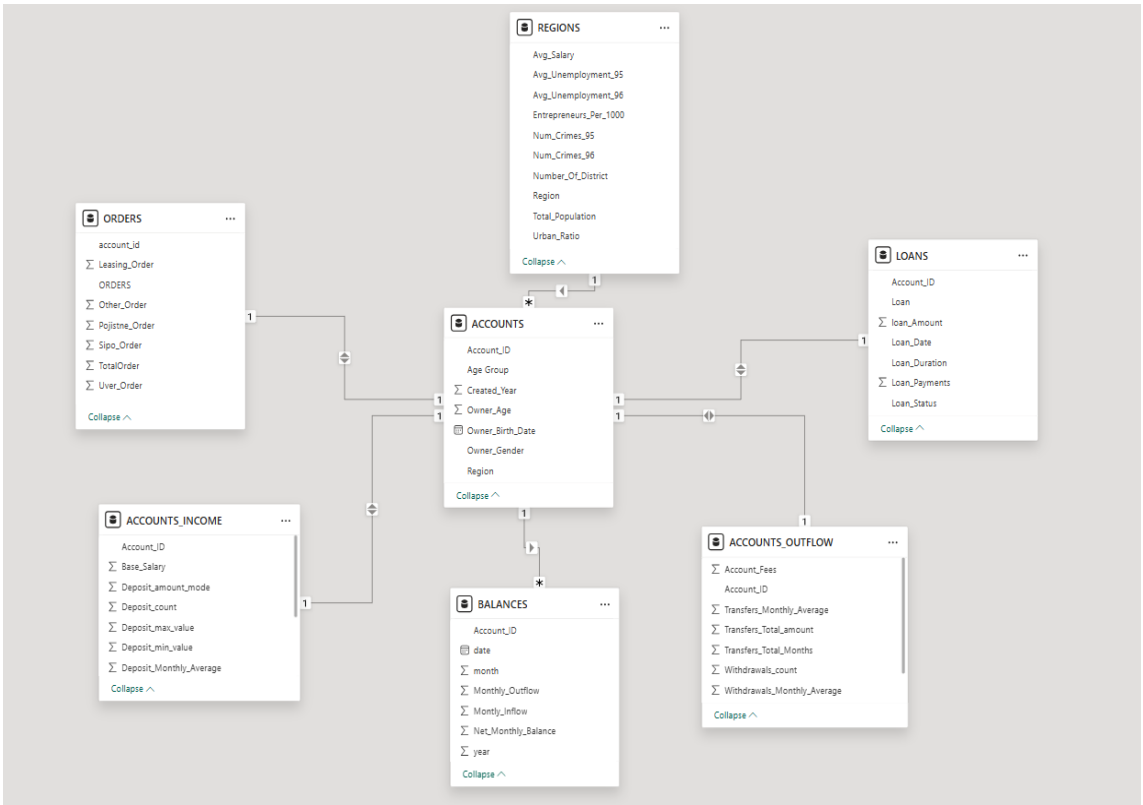
The data was accessed by connecting to a public academic server hosted by Czech Technical University in Prague [relational.fel.cvut.cz] via MySQL Workbench, from which the required data was extracted and transformed using SQL [see SQL_Scripts file]. The result was 7 clean, structured tables exported as CSV files [see Data folder] - ready for analysis.

The modeling process started from the business itself. The account was identified as the core entity - it is the point where every relationship between the bank and its clients exists. Everything else was built around it.

For each account, the model captures the owner's demographics (gender, age, region), financial behavior (incoming, outgoing, balance) and product relationships (loans, orders). The result is a complete profile of the bank's activity: not isolated tables, but a connected whole centered on the account.



Database structure

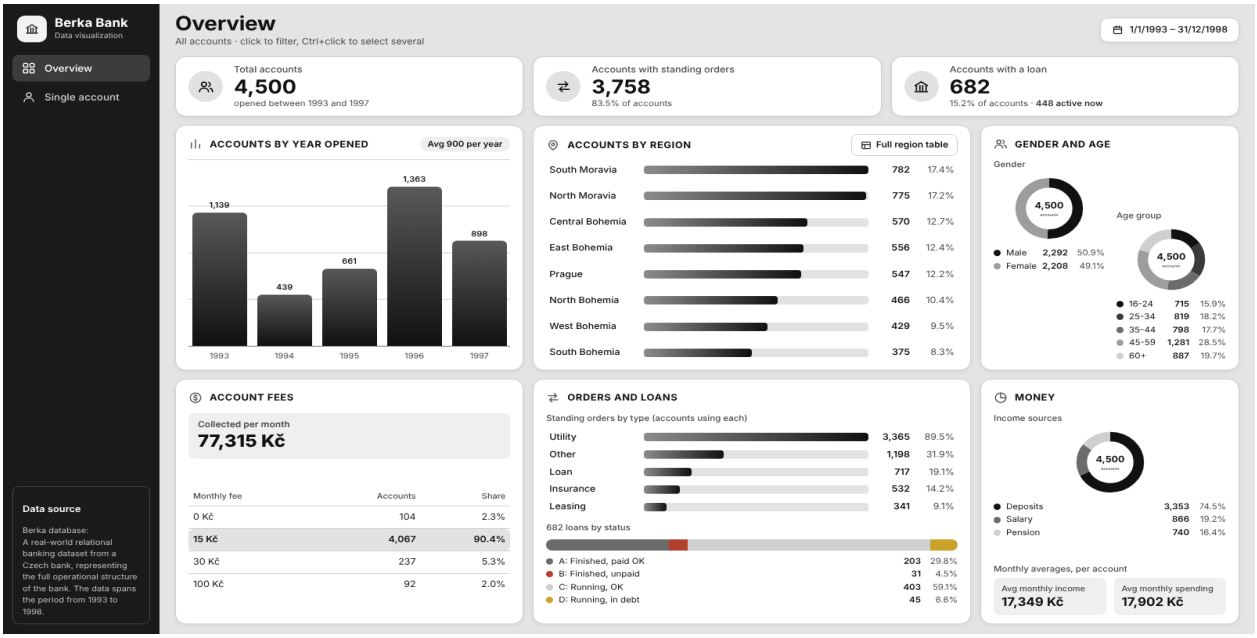


Extracted data structure

3. Deliverable 1: Interactive Visualization

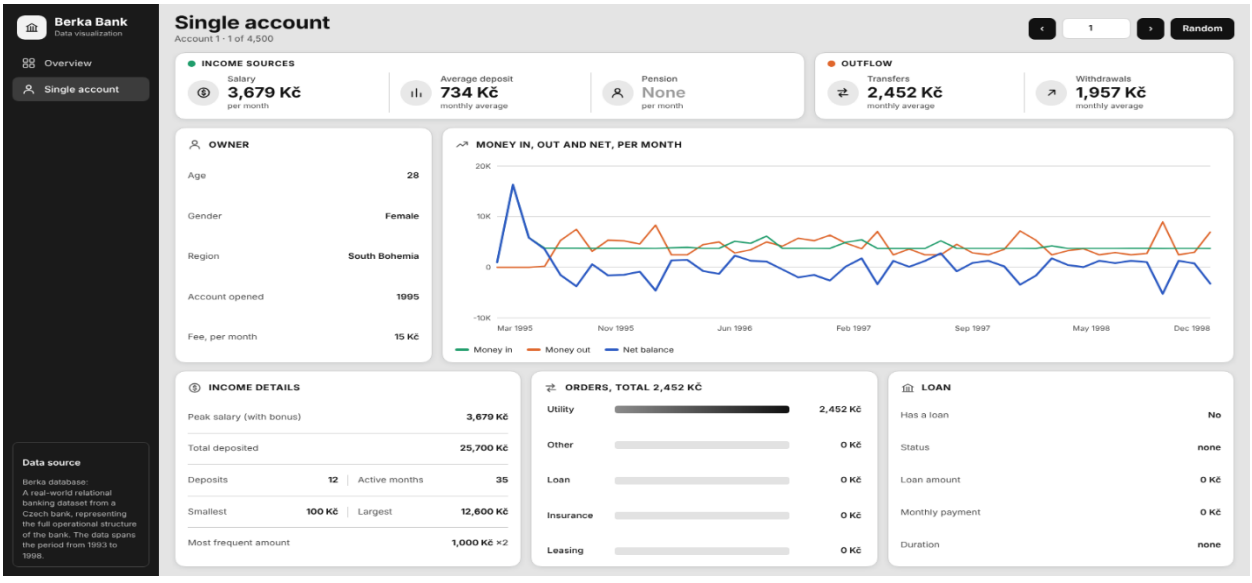
A two-page visualization built on one master table with one row per account using plain HTML, CSS, JS. Click any bar, slice or row to filter the whole page; Ctrl+click compares several values, such as two regions.

Overview. The whole bank at a glance: accounts by year, region, gender and age, plus fees, standing orders, loans and income sources.



Overview

Single account. The full profile of one account: income, outflow, deposits, orders, loan, and its monthly money in, money out and net balance.



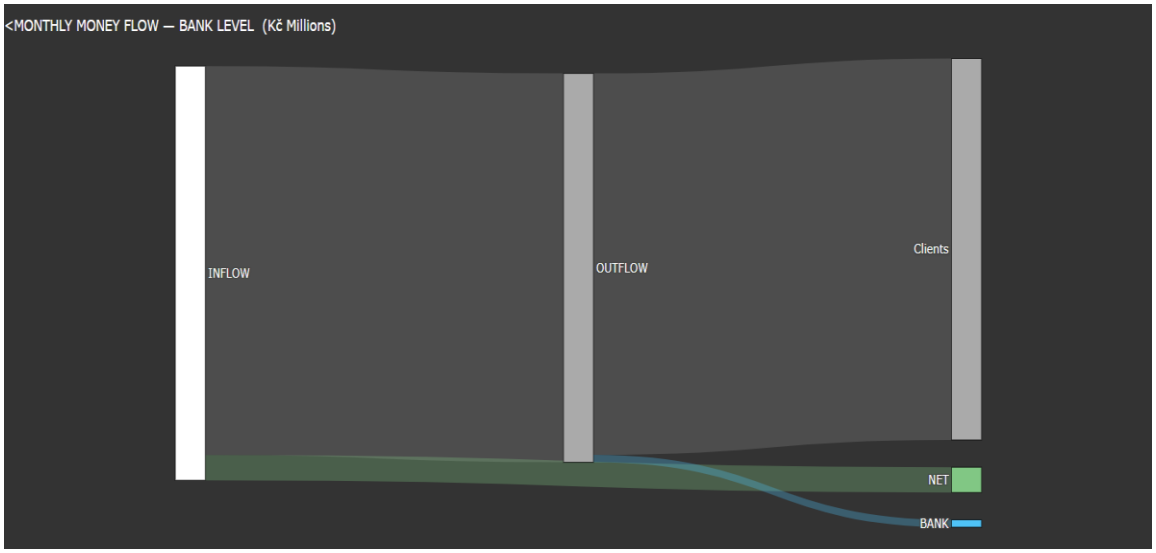
Single account

4. Deliverable 2: Bank-level Money Flow

Reconstruct the bank's approximate monthly cash flow by standardizing its full transaction history (trans table) to a monthly average.

The history is standardised to an average month at bank level: inflow (PRIJEM), outflow (VYDAJ, VYBER), fees (SLUZBY) and loan repayments (UVER), each divided by the number of months.

Item	Kč per month	Share
Total inflow	44,826,148	100% of inflow
Total outflow	42,088,605	93.9% of inflow
Client outflows	41,283,090	98.1% of outflow
Fees	38,112	0.1% of outflow
Loan repayments	767,403	1.8% of outflow
Net (remaining balance)	2,737,543	6.1% of inflow



Monthly money flow, bank level (Kč millions)

5. Key Observations

From the interactive visualization pages

- The bank hold 4,500 accounts, opened between 1993 and 1997. Openings are inconsistence: peaks and drops (1,363 in 1996, and a 439 in 1994).

- South Moravia and North Moravia hold the most accounts, together 35% of all bank accounts. But they are the lowest compared to others in accounts per 1,000 residents (South Bohemia is the highest per 1,000 residents, but it is also the lowest in account share). Prague, with 12% of accounts, has the highest average salary and the lowest unemployment of all regions.

- Clients are split almost evenly by gender. The largest age group is 45–59 (29%); each of the other four groups holds 16–20%.

- Almost every account (90% of accounts) pays a small monthly fee: 15 Kč. Total fees collected per month are about 77K Kč.

- 3,758 accounts (84%) have standing orders, and utility payments dominate them (90% of those accounts).

- Only 682 accounts (15%) have a loan. Most are healthy: 89% are in good standing.

- An average account brings in about 17,300 Kč per month and sends out about 17,900 Kč. Deposits are the main source of income (3,353 accounts); salary reaches 866 accounts and pension 740.

From the money flow

- In an average month, 44.8M Kč enters the bank and 42.1M Kč leaves it. About 2.7M Kč (6% of inflow) stays.

- 98% of what leaves is client activity. Only 1.9% (about 0.8M Kč) goes to the bank itself, mostly loan repayments; fees are just 38K Kč.

In short: a stable, active client base that moves large amounts of money every month, while the bank keeps little of it directly. Fees are tiny and only one account in seven has a loan.